RPMGLOBAL

AMT - Equipment Management Issues

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical, hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

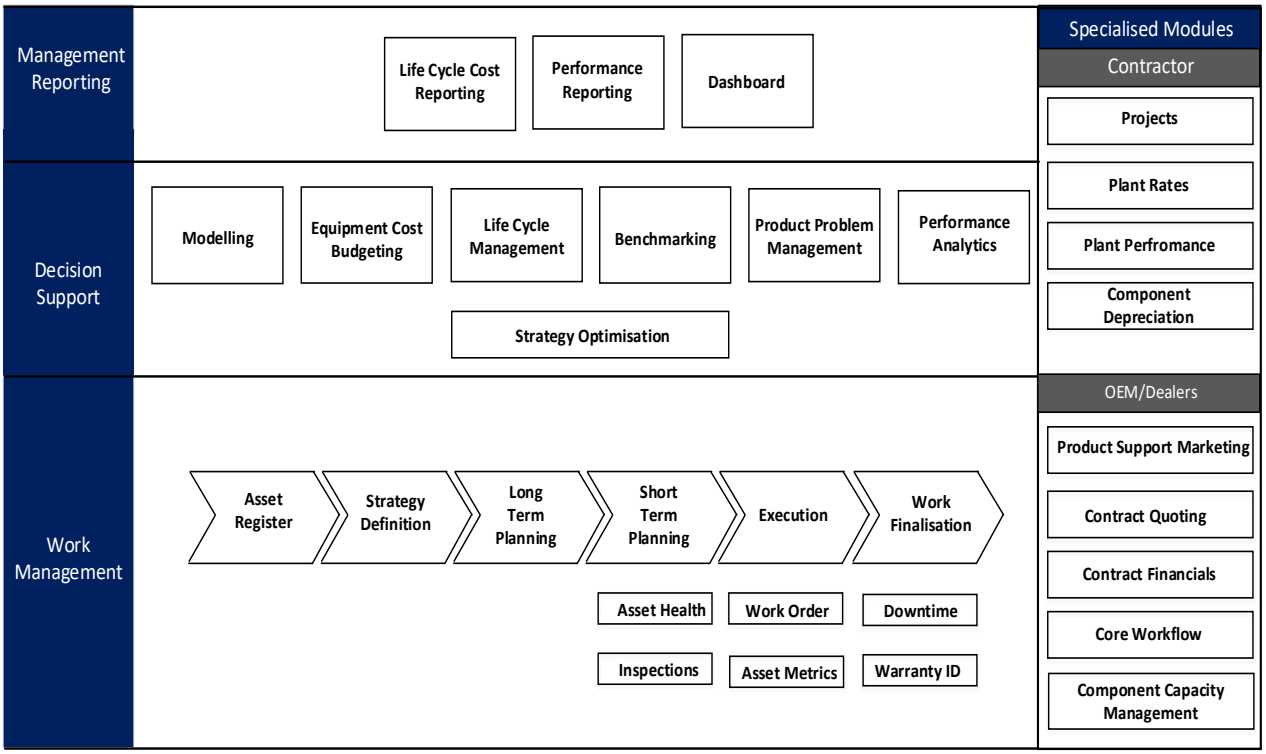
Check your understanding on the workshop and ask any related questions.

Course Outline

Prerequisites	
Target Audience	Senior Management, Maintenance Manager, Contract Manager.
Nominal Duration	
Course Objectives	
Learning Outcomes	At the end of the course, participants will have learned to: • Add, evaluate, and monitor equipment management issues, as well as associated tasks.
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below. The functionality covered by this training manual is identified by the Strategy Optimisation section in the diagram.



OVERVIEW

EM issues is short for equipment management issues, which covers all issues directly related to an equipment type or model. Issues include product problems (equipment design, equipment manufacture), OEM updates, continuous improvement initiatives, significant application and operation issues, poor maintenance practices (redo, inappropriate workscopes, and so on), and general performance items that need to be addressed.

Training Workshops

1. Equipment Management Issue Process

Procedure – Registration

After an issue directly related to a model of equipment has been identified, it should be registered as an equipment management issue (EMI) in AMT. EMI manages all model-specific issues, not just the traditional product problems. For example, if an equipment operator is over-braking, causing early failures and thereby affecting costs and availability, this should be registered as an EMI. The root cause is not the OEM, rather it is related to the customer's operations, which are affecting equipment, costs and availability. Therefore, the issue should be registered in order that it receives attention.

1. Navigate to **Knowledge Database** > **EM Issues** > **EM Issues**. This opens your web browser.
2. Click **Add** and complete the fields.



Enter a resolution date to monitor the EMI through the process. If the EMI passes the resolution date, it will be flagged as overdue.

3. When all the relevant information regarding the EMI has been entered, select **Registration Complete**. AMT moves the EMI into the next phase of the process – Technical Investigation.

Procedure – Technical Investigation

A technical analysis is performed to discover and describe in detail the symptoms and causes of the EMI. Documents that support the analysis should be attached to the EMI.

1. To add technical information to an EMI, navigate to **EM Issues**.
2. Search for the EMI using the filters; notice the stage of the EMI and whether the EMI is overdue. Click **Edit** to open the EMI, and enter the appropriate details.
3. When all the relevant technical information has been entered, select **Analysis Complete**. AMT moves the EMI into the next phase of the process – Evaluation.

Procedure – Evaluation

The evaluation of an EMI must be objective. The evaluation should be based on the EMI's impact on the key drivers of equipment management, that is, equipment design, application and operation, and operational efficiency (strategy, planning, daily operations, administration and customer service). The impact is typically quantified in terms of the number of equipment affected by the EMI and resulting cost and availability side effects.

1. To add evaluation information to an EMI, navigate to **EM Issues**.
2. Search for the EMI using the filters; notice the stage of the EMI and whether the EMI is overdue. Click **Edit** to open the EMI, and enter the appropriate details (see linking EM issues to equipment, costs and availability later in this document).
3. When all relevant evaluation information has been entered, select **Evaluation Complete**. AMT moves the EMI into the next phase of the process – Authorisation.

Procedure – Authorisation

The potential outcomes of the evaluations of EMI are reprioritising, changing the status, and possibly recommending further investigation, which is submitted for authorisation.

1. To add authorisation information to an EMI, navigate to **EM Issues**.
2. Search for the EMI using the filters; notice the stage of the EMI and whether it is overdue. Click **Edit** to open the EMI, and enter the appropriate details.
3. If approval was given to continue the investigation of the EMI, select **Authorised – Yes**. AMT moves the EMI into the next phase of the process – Investigation.
4. If approval was not given, select **Authorised – No**. AMT changes the EMI's status to **Abandoned**.

Procedure – Short Term Action

1. To add a short-term action to an EMI, navigate to **EM Issues**.
2. Search for the EMI using the filters; notice the stage of the EMI and whether it is overdue. Click **Edit** to open the EMI, click **Short Term Action**, click **Edit**, and then select **Is Short Term Corrective Action Applicable?** if there is interim containment action needed to resolve the issue.
3. When all the relevant action information has been entered, select **Short Term Action Complete**. AMT moves the EMI into Long Term Action.



*If the short-term action requires a backlog work order to be created and sent to a planner, click **Create Backlogs** (refer to the section after Long Term Action for more details).*

Procedure – Long Term Action

1. To add a long-term action to an EMI, navigate to **EM Issues**.
2. Search for the EMI using the filters; notice the stage of the EMI and whether it is overdue. Click **Edit** to open the EMI, click **Long Term Action**.
3. When all the relevant action information has been entered, select **Long Term Action Complete**. AMT moves the EMI into the next phase in the process – Monitor and Resolution.



*If the long-term action requires a backlog work order to be created and sent to a planner, click **Create Backlogs** (refer to section below).*

Procedure – Work Order Settlements

1. You can enter summary costs for performing the short and long term actions into an EM Issue by selecting the **Workorder Settlements** menu option.



These work order settlements do not look up any linked AMT work order settlements. The costs entered are for reference only.

Procedure – Creating a Backlog from a Short/Long Term Action

One of the advantages of using EM issues in AMT is that action plans can be directly communicated from EM issues into planning for scheduling and equipment status for execution (that is, it's the one system). The example below illustrates the process for implementing a long-term action in EM Issue, but note that a short-term action follows the same process.

In the long-term action, the Repair Details section is actually a backlog template that you complete. However, you don't have to create a backlog for the planner if it is not required.

After the repair details have been completed, you must click **Save**. A new option displays in the **EM Issue** menu called **Create Backlogs**.

EM ISSUES – LONG TERM ACTION EXAMPLE



1. To create backlogs for the long-term action, click **Create Backlogs** on the menu.
2. The Backlog Workorder Creator screen opens. From here, you can filter for the equipment to which the long-term action is to be applied.
3. Select **Create Workorder** next to the equipment to which the long-term action is to be applied. When you're done, click **Save**.

   AMT then begins the process of adding the EM issue's long-term action backlog to the equipment you selected.

   The new backlogs will appear in the planning and equipment status modules of AMT.

Procedure – Monitor and Resolution

A follow-up usually occurs after a short-term or long-term action has been implemented. It is a check to ensure that the action has been implemented correctly (as per specification) and that the implemented action has had the desired effect. It could be dangerous to assume that the action will work perfectly first time. Proactive checks, tests (whatever is necessary) should be performed to verify the quality of the implemented action, and whether the action has indeed resolved the issue. After all parties are satisfied that the EMI has been resolved, it should be closed.

1. To enter resolution information to an EMI, navigate to **EM Issues**.
2. Search for the EMI using the filters; notice the stage of the EMI and whether it is overdue, and click **Resolution**. Complete resolutions notes, adjust the closure date, and add any follow-up check information.



You only need to complete the abandon notes if you want to abandon the issue.

3. When all the relevant action information has been entered, select **Save**. The EMI will be set to closed or abandoned status.

Learning Check



Check your understanding of the workshop and ask any related questions.

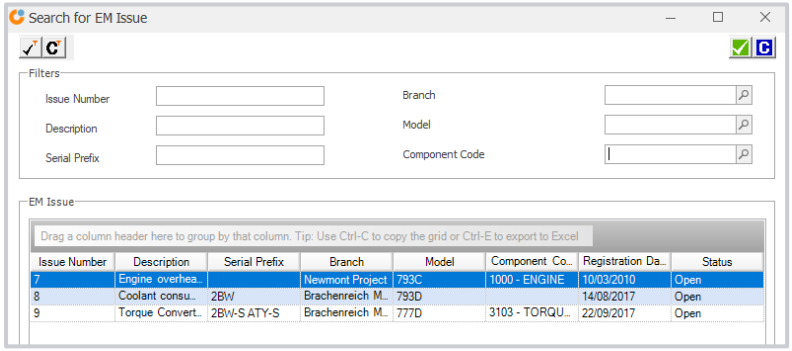
Notes

2. Linking Maintenance History to EM Issues

Procedure – Link Equipment

1. The quantity of equipment affected is established by linking an EM issue to events and work orders across one or more equipment. Navigate to **EM Issues**.
2. Click the **Workorders** tab. Search for the relevant work order and open it. On the **Feedback** tab of the work order, click **Linked EM Issue** in **Other Details** tab.

LINK EM ISSUE SCREEN



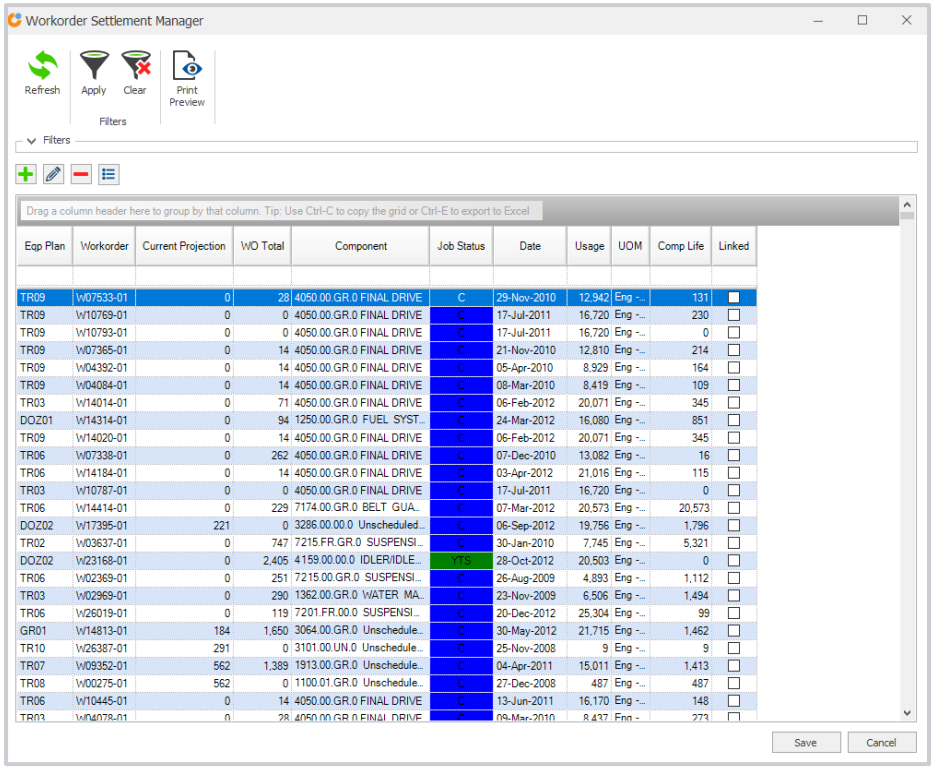
3. Search for the EMI and double-click it to link.

Procedure – Link Costs

The cost impact of the EMI is quantified by linking a work order (that is already linked to the EMI) to a work order settlement.

1. Navigate to **EM Issues**.
2. Click the **Workorders** tab. Search for the relevant work order, open, and click the **Settlement** section of the specific work order in the event.
3. If the work order is linked to a work order settlement, the work order settlement number and/or <DUE> is displayed (<DUE> means that AMT is waiting for the work order settlement from the ERP system). If a work order settlement number or <DUE> is not displayed, and the work order has been completed, search for the work order settlement by clicking **Link to WO**.

LINK WORKORDER SETTLEMENT SCREEN



4. Search for the work order settlement and link it to the work order. Consequently, since the work order is already linked to the EMI, the costs from the work order settlement will be associated with the EMI.



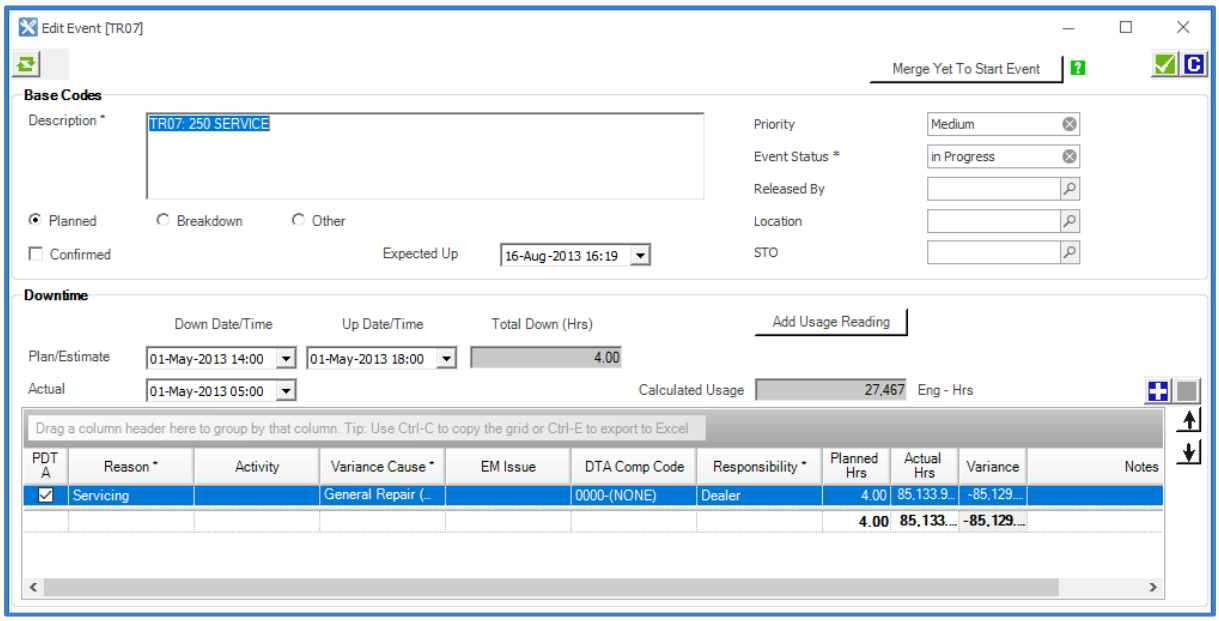
If the work order settlement cannot be found, contact the System Administrator.

Procedure – Link Downtime (Availability Impact)

Identifying the downtime associated with the EMI quantifies the impact on availability. This is done in equipment status by recording how much downtime during an event was related directly to the EMI.

1. Under **Daily Planning**, double-click **Planning**.
2. Click the **Events** tab. Search for the relevant work order and open the event (not the work order). Scan the downtime allocations, and identify which portion of the downtime was relevant to the execution of the EMI. If multiple work orders were performed during the downtime, but only one downtime allocation for maintenance exists, create an additional downtime allocation for maintenance that specifically relates to the EMIs.

DOWNTIME ALLOCATION EXAMPLE



3. Click **EM Issue**, locate the EMI and click **Save**.

Procedure – Results of Linking

1. Navigate to **EM Issues**.
2. In the EM Issues module, search for the EMI and click **Preview/Print** to get a summary of the current position. Notice the updates under the Impact section of the report.

EM ISSUE SUMMARY

Equipment Management Issue Summary

Issue Number	**11**	Issue Status	**Over Due**
Issue Type	**Defect**	Issue Stage	**Technical Analysis**
Model	**MD831**	Priority	**Medium**
Registered Date	**07-Nov-2003**	Branch	**Gold Mine**
Required Date	**07-Jan-2004**	Safety Issue	
		Authorised to Proceed	
Description			

Registration Details

Serial Prefix	**9YF**	Component Code	**3101 - TORQUE CONVERTER**
Serial Numbers	**ALL**	Modifier Code	
		Symptom Code	**4 - Low Power**

Evaluation

	Qty Equipment	Cost Impact	Availability Impact
MARC			
NON MARC			
Total	0	0	0%

Component Life

Expected Component Life		Average Component Life	

Impact

Number of Equipment	**1**	Downtime Events	**1**
Number Of Tasks	**1**	Downtime Hours	**10.00**
Total Cost of Tasks	**60,469.82**		

*The impact is quantified in under the **Impact** heading.*

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes